

Answer: A

- 23** For the magnet to fall slower, the induced upward magnetic force on the magnet has to increase, or the resultant downward force on the magnet has to decrease.

Option A: Releasing the magnet from a smaller height will cause the rate of change of magnetic flux through the pipe to decrease since the magnet enters the pipe with a lower speed. Hence, the induced e.m.f. in the pipe will be of a smaller magnitude, which will result in a smaller current and smaller induced magnetic force.

Option B: A pipe with a higher resistivity will cause the induced current to be of a smaller magnitude, which will result in a smaller induced magnetic force.

Option C: A weaker magnet will cause the induced e.m.f. to be of a smaller magnitude, since the rate of change of magnetic flux through the pipe will decrease.

Option D: The induced magnetic force is the same and the resultant downward force will decrease due to the smaller weight of the magnet.

Answer: D

$$24. \quad I = \frac{\sqrt{(5.0)^2(1) + (3.0)^2(1)}}{1} = 4.1 \text{ A}$$